Manish Desireddi

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LinkedIn | GitHub | Portfolio

EDUCATION

KL University — Vijayawada, India
B.Tech - Artificial Intelligence & Data Science — 2022 – 2026
CGPA: 8.65 / 10.0

TECHNICAL SKILLS

Languages: Python, C++, C#, JavaScript, TypeScript, SQL, Bash.
AI/ML & Frameworks: PyTorch, TensorFlow, LangChain, LangGraph, HuggingFace, FastAPI, React.js, Next.js.
Infrastructure & Tools: Docker, Kubernetes, Apache Airflow, Jenkins, Git, AWS, Azure, Linux.
Databases & Brokers: PostgreSQL, MongoDB, VectorDB, Redis, Celery, RabbitMQ.
Core Engineering: System Design, Data Structures & Algorithms, MLOps, CI/CD, CUDA, ROCm, Selenium.

EXPERIENCE

Nokia Solutions and Networks — Bengaluru, India
Student Intern — Aug 2025 – Jun 2026

- Automated repetitive network UI testing across ~80 operational scenarios by engineering robust Python Selenium scripts, making sure strict <4GB memory utilization thresholds and preventing manual regression faults.
- Streamlined continuous operational stability monitoring by developing Apache Airflow ETL pipelines, successfully extracting and ingesting dynamic performance telemetry metrics into Elasticsearch to reduce manual KPI verification cycles.
- Eliminated duplicate configuration value violations across network base stations by designing a dynamic Python automation script, actively reducing XML generation overhead and ensuring foundational RAML compliance.
- Engineered Python-based utility modules to support a scalable AI test automation environment, integrating workflows with Jenkins and Robot Framework to systematically accelerate release cycles.

TECHNICAL PROJECTS

Sub-Quadratic Audio Transformer Engine | *C++, CUDA, PyTorch, PyBind11, CMake*

- Engineered a bare-metal, lock-free linear attention engine bypassing standard $O(N^2)$ memory barriers, achieving $O(N)$ memory causal attention and >200k tok/s throughput to enable infinite continuous audio streaming.
- Orchestrated zero-copy polymorphic tensor interoperability by integrating custom CUDA/HIP macros with PyBind11, cross-compiling exact PyTorch TRL integrations to AMD ROCm and NVIDIA hardware while maintaining a <256MB VRAM footprint.

Self-Healing UI Automation Engine | *Python, Selenium, ONNX, FAISS*

- Architected an intent-free self-healing UI locator utilizing ONNX embeddings, autonomously resolving broken Selenium tests in real-time as measured by a highly accurate 7-factor weighted scoring ensemble.
- Intercepted failed element calls by engineering a custom fallback mechanism using BM25 and structural tree edit distance, dynamically matching parsed selectors against a growing DOM baseline to rewrite test scripts on disk.

Agentic Commerce Platform | *Python, LangGraph, FAISS, Redis, PostgreSQL*

- Developed a multi-tier autonomous commerce architecture orchestrating 15 specialized active agents via LangGraph, executing dynamic Dual-RAG semantic searches and continuous ReAct tool-calling loops over Redis.
- Engineered Agent-to-Agent (A2A) negotiation and post-purchase reverse logistics via JSON-RPC, establishing robust, fail-forward task execution utilizing Celery and cryptographic idempotency protocols.

Agentic Desktop AI Assistant - Cloud | *FastAPI, LangGraph, ChromaDB, Tauri*

- Built a low-latency Agentic Desktop Assistant utilizing FastAPI and 6-provider LLM routing, delivering sub-500ms first-token latency and priority failover while maintaining a highly efficient 300MB zero-GPU cloud footprint.
- Hardened the application's tool execution surface by deploying SQLite-checkpointed sessions, SSRF-validated networking, and hash-deterministic fallback embeddings via Gemini 2.0 and ChromaDB.